

Piping & Instrumentation Description

Document No: BPL-PID-CDU-001 | System: Crude Distillation Unit

Process Overview

Raw crude oil from storage is pumped via P-101A/B through a series of pre-heat exchangers (E-101 to E-104) before entering the Desalter (D-101) to remove salts. The desalted crude is further heated and sent to the F-101 Furnace, where it is partially vaporized before entering the Atmospheric Column C-101.

Control Loops & Interlocks

FIC-101: Controls raw crude flow rate. Cascaded with level controller LIC-201 on the desalter.

TIC-105: Controls furnace outlet temperature by modulating fuel gas control valve TV-105A.

Interlock I-101: On low crude flow (FAL-101), trip furnace F-101 to prevent tube coking.